

Day 67 – DNS & Name Resolution

Overview

Day 67 introduced **DNS (Domain Name System)** and its role in modern network communication. While IP addresses are used by devices to communicate, humans prefer using easy-to-remember names such as websites and server hostnames. DNS provides the translation between hostnames and IP addresses.

Building on the previous labs:

- Day 65 introduced RIP dynamic routing.
- Day 66 introduced DHCP automatic address assignment.
- Day 67 introduced DNS name resolution.

The objective of this lab was to configure a DNS server, create DNS records, distribute DNS information through DHCP, and test hostname resolution across the multi-router network.

Objectives

By the end of this lab, I was able to:

- Understand the purpose of DNS.
 - Understand how DNS translates hostnames into IP addresses.
 - Configure a server with a static IP address.
 - Configure and enable DNS services.
 - Create DNS records.
 - Distribute DNS server information through DHCP.
 - Integrate DHCP and DNS services within an existing RIP network.
 - Practice troubleshooting name resolution issues.
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DNS Fundamentals

What is DNS?

DNS stands for:

Domain Name System

DNS translates:

Human-readable names

into:

IP addresses

Example:

server.local

becomes:

192.168.30.100

Without DNS, users would need to remember IP addresses for every service.

Why DNS Matters

DNS is used everywhere:

Enterprise Networks

- Internal servers
- File shares
- Printers
- Applications

Cloud Platforms

- AWS Route 53
- Azure DNS
- Google Cloud DNS

Home Networks

- Websites
 - Streaming services
 - Online gaming
-

Relationship to Previous Labs

The network now includes three major services:

RIP

Moves packets between networks.

DHCP

Automatically assigns IP addresses.

DNS

Resolves names into IP addresses.

Together these services form the foundation of enterprise networking.

DORA Review from Day 66

Before DNS can work:

DHCP provides:

- IP Address
- Subnet Mask
- Default Gateway

→ DNS Server

This means DNS depends on DHCP configuration.

Network Topology

LAN10 LAN20 LAN30

PC0 PC1 PC2

|||

SW0 SW1 SW2

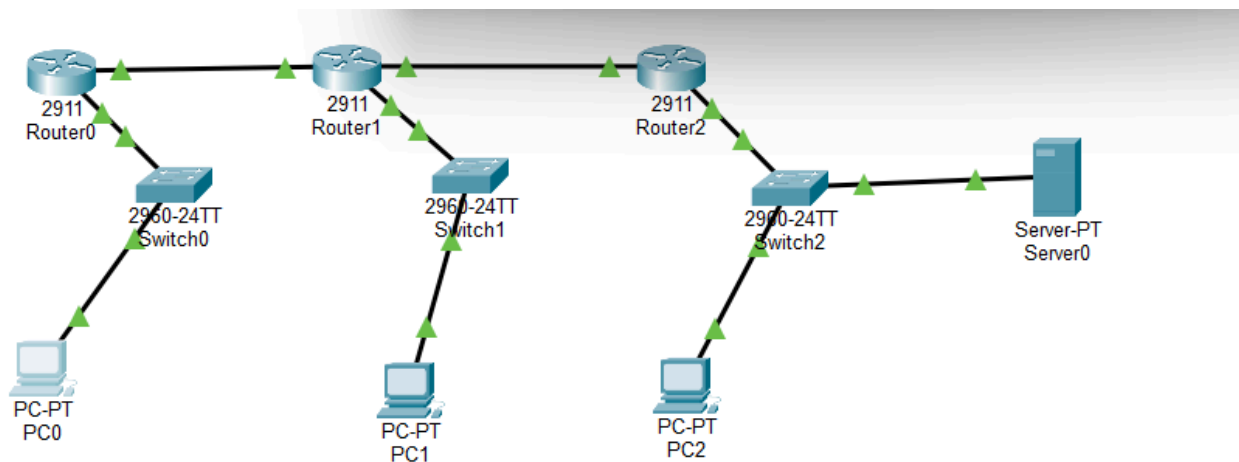
|||

R0 ----- R1 ----- R2

|

Server0

Screenshot #1 – Day 67 Topology



Highlight

- Three routers
- Three switches

- Three PCs
 - Server0 added to LAN30
 - RIP routing active
 - DHCP active
-

DNS Server Configuration

Server Configuration

Server0 was assigned a static address.

IP Address:

192.168.30.100

Subnet Mask:

255.255.255.192

Default Gateway:

192.168.30.1

DNS Server:

192.168.30.100

Screenshot #2 – Server0 IP Configuration

The screenshot shows a network configuration window titled "IP Configuration". It has a blue header bar. Below the header, there's a section for "IP Configuration" with two radio buttons: "DHCP" (unselected) and "Static" (selected). To the right of the "Static" radio button is a blue circle icon. Below this, there are four rows of configuration fields, each with a label on the left and a value on the right:

IPv4 Address	192.168.30.100
Subnet Mask	255.255.255.192
Default Gateway	192.168.30.1
DNS Server	192.168.30.100

Highlight

Static Address

192.168.30.100

DNS Service Configuration

DNS Service was enabled on Server0.

A DNS record was created.

Example:

Name:

server.local

Address:

192.168.30.100

Screenshot #3 – DNS Service Configuration

The screenshot shows a web-based configuration interface for DNS. At the top, there's a toggle for 'DNS Service' which is currently 'On'. Below this, the 'Resource Records' section is visible. It includes a 'Name' field with 'server.local', a 'Type' dropdown menu set to 'A Record', and an 'Address' field with '192.168.30.100'. There are 'Add', 'Save', and 'Remove' buttons. Below the form is a table with columns: No., Name, Type, and Detail. The table contains one entry with '0' in the No. column, 'server.local' in the Name column, 'A Record' in the Type column, and '192.168.30.100' in the Detail column.

No.	Name	Type	Detail
0	server.local	A Record	192.168.30.100

Highlight

DNS Service = ON

server.local

192.168.30.100

DHCP Integration

The DHCP pools were modified so clients would automatically receive the DNS server address.

Example:

```
ip dhcp pool LAN10  
dns-server 192.168.30.100
```

The same modification was applied to:

LAN20
LAN30

Client Verification

Client systems successfully received:

DNS Server:
192.168.30.100

through DHCP.

Example:

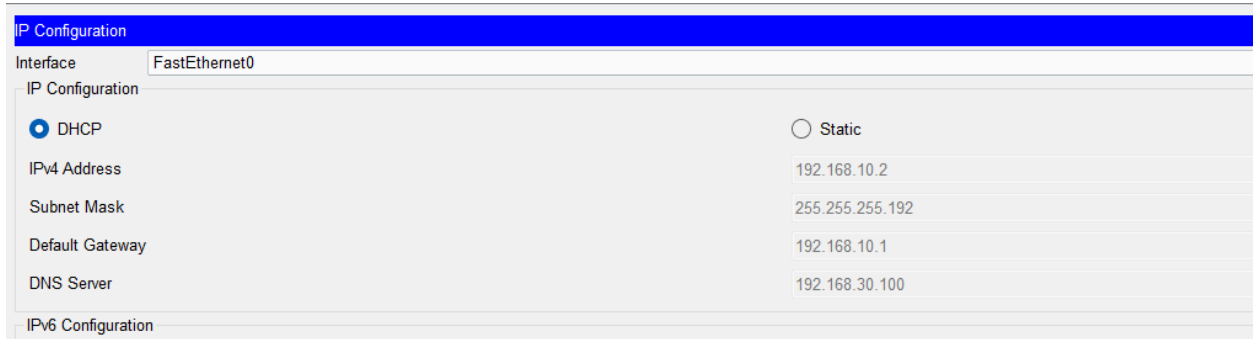
PC0

IP:
192.168.10.2

Gateway:
192.168.10.1

DNS:
192.168.30.100

Screenshot #5 – PC0 DHCP Information



The screenshot shows the 'IP Configuration' window for the 'FastEthernet0' interface. The 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The configuration details are as follows:

Field	Value
IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.1
DNS Server	192.168.30.100

The 'IPv6 Configuration' section is visible at the bottom but is not expanded.

 INSERT SCREENSHOT HERE

Highlight

DNS Server
192.168.30.100

Testing

Connectivity Testing

The network infrastructure remained operational.

DHCP continued functioning correctly.

RIP routes continued functioning correctly.

Multiple LAN-to-LAN ping tests succeeded.

DNS Testing

Attempted:

ping server.local

Result:

Name resolution unsuccessful

during the lab.

Further troubleshooting revealed communication issues between Server0 and Router2 despite the server being configured with the correct static addressing.

Troubleshooting Performed

The following items were verified:

- ✓ Static IP configuration
- ✓ Correct subnet mask
- ✓ Correct default gateway
- ✓ DHCP DNS distribution
- ✓ Green switch link indicators
- ✓ Server interface enabled
- ✓ RIP routing operational

The DNS implementation was partially completed, but full name resolution was not achieved during the lab session.

This troubleshooting experience helped reinforce the importance of validating lower-layer connectivity before assuming DNS configuration issues.

Key Concepts Learned

Static vs Dynamic Addressing

PCs

Use DHCP:

Automatic assignment

Servers

Use Static IPs:

Permanent address

Reason:

Clients must always know where the DNS server lives.

DNS Resolution Flow

Client

↓

DNS Query

↓

DNS Server

↓

IP Address Response

↓

Connection Established

What I Learned

- DNS converts names into IP addresses.
- DNS is used by cloud providers and enterprise environments.
- DNS servers commonly use static addressing.
- DHCP can distribute DNS server information automatically.
- Name resolution depends on underlying network connectivity.

- DNS troubleshooting often begins with basic ping testing.
 - Services build on top of existing networking infrastructure.
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Reflection

Day 67 introduced DNS and demonstrated how network services work together. I configured a DNS server with a static IP address and integrated DNS information into the existing DHCP infrastructure. Although full hostname resolution was not successfully completed during the lab, the exercise provided valuable troubleshooting experience and reinforced the relationship between DHCP, DNS, and routing services. This lesson expanded the network beyond simple connectivity and moved toward enterprise-style service delivery.